Robustness of a Bayesian First Order AutoRegressive Model under Prior Perturbations

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February 2026

1 Statement of the Quantitative Finance Problem

1.1 Background and Motivation

Volatility modelling is a ubiquitous problem in financial engineering, underpinning applications in financial risk management, asset pricing, and portfolio optimisation. It involves measuring, estimating, and forecasting a range of stylised features that characterise the temporal evolution of asset returns. Ongoing research in this area reflects both the economic importance of volatility and the heterogeneity of its behaviour across asset classes and time scales. For example, Gao et al. (2024) investigate high-frequency equity volatility using a latent GARCH–Itô Bayesian framework, while Zhao et al. (2024) examine daily volatility forecasting by extending GARCH models with neural network architectures. These contrasting approaches highlight that volatility dynamics, and the modelling tools required to capture them, vary substantially with data frequency and market context.

An effective volatility model must account for several empirically observed stylised facts. These include, but not limited to, persistence of volatility, mean reversion and clustering of volatility (Engle and Patton (2001)). Failure to capture these features can lead to systematically biased risk measures, mispriced derivatives, and suboptimal portfolio decisions.

A wide range of econometric models has been proposed to model these stylised facts, including Autoregressive Moving Average (ARMA) models, Autoregressive Conditional Heteroscedasticity (ARCH) and GARCH frameworks, stochastic volatility models, regime-switching models, and threshold-based approaches (Xiao, Aydemir (2007)). While each class of models captures particular aspects of volatility behaviour, the presence of multiple interacting stylised features complicates both modelling and forecasting. As noted by Gallo and Otranto (2015), linear specifications often struggle to accommodate such complexity, motivating the development of non-linear and regime-based models across a range of asset classes and sampling frequencies (Tan, Wu (2025), Cheikh et al. (2020), Luo et al. (2022)).

Within this context, stochastic processes and regime-switching frameworks have emerged as natural tools for volatility modelling. Ding, Kambouroudis, and McMillan (2025) compare a linear Heterogeneous AutoRegressive (HAR) model with a regime-switching Markovian specification and find superior forecasting performance under the latter, reflecting the ability of regime-based models to accommodate structural changes in volatility dynamics. More broadly, regime-switching models such as Hidden Markov Models (HMMs) offer a flexible representation of volatility behaviour by allowing distinct latent states with differing persistence and variability, thereby capturing key stylised features that are difficult to model within a single-regime framework. As the adoption of non-linear models has increased, the use of stochastic processes has become increasingly important in volatility modelling. Despite these advances, a persistent challenge remains: determining which non-linear modelling frameworks are most appropriate for practical use, and under what conditions.

This adoption from linear conditional variance models to stochastic and regime-based formulations reflects a broader shift in quantitative finance toward modelling volatility as an unobserved, dynamically evolving state rather than a deterministic function of past returns. This shift allows richer representations of market behaviour, but at the cost of increased reliance on complex estimation techniques whose inferential reliability may be suboptimal. It follows that these developments motivate a shift in focus from model construction alone to research into the reliability of the posterior distributions produced by advanced estimation methods. In Bayesian stochastic volatility and regime-switching models, posterior inferences form the basis for risk assessment, forecasting, and decision-making. Consequently, understanding when such posterior distributions genuinely reflect underlying volatility dynamics—and when they arise as artefacts of modelling assumptions or parameter instability—has become a critical issue. This study is therefore motivated by the need to systematically analyse the robustness of Bayesian SV-HMM posterior inferences when modelling complex, non-linear volatility time series.

1.2 Statement of the problem

Bayesian inferences have the capacity to excel at volatility modelling, however a central challenge is determining the economic and statistical robustness of the posteriors calculated as they may be reflections of modelling assumptions, regime instability and poor parameter identification rather than the underlying market behaviour.

This issue is highlighted in stochastic volatility modelling settings where Bayesian estimation methods are necessary due to the intractability inherent to stochastic processes when calculating likelihoods, yet may estimate posterior distributions that are unstable and poorly reflect market volatility characteristics. Therefore, in-depth and comprehensive posterior analysis is necessary to investigate the reason behind the obtained posterior distribution to determine the utility of results.

This study implements a Bayesian Autoregressive model where parameters are inferred via the No-U-Turn Sampler (NUTS) MCMC method to obtain a posterior distribution to model volatility in the form of log variance of returns. A distribution evaluation framework, Kullback-Leibler (KL) divergence, paired with predictive performance measure Expected Log Pointwise Predictive Density (ELPD), are employed to assess the stability and predictive power of the estimated posterior distributions. In order to evaluate the robustness of the posterior distribution obtained, Prior Perturbation techniques are employed. By perturbing the prior inputs into NUTS, variations in KL divergence and ELPD statistics will create a holistic picture in distinguishing whether the initial posterior distribution robustly obtained captured a time series' volatility factors or if it's a result of modelling assumptions and regime instability.

1.3 Scope, Limitations and Assumptions

The scope of this research study is a posterior validation study, revolving around evaluating the robustness and reliability of posterior distributions obtained from an Autoregressive model estimated via NUTS in Python across the Equity and Foreign Exchange markets. Obtained posterior distributions are used to model volatility, followed by implementation of prior perturbation techniques to obtain new distributions. New distributions are evaluated with via KL Divergence and ELPD to assess their reliability to determine whether results are artifacts of modelling assumptions or regime instability from the model or if stylized factors are captured robustly.

Modelling assumptions. Returns are assumed to be conditionally Student-t distributed given latent volatility and regime states. Regime dynamics are assumed to follow a first-order Markov process with time-homogeneous transition probabilities. Prior distributions are specified to be weakly informative, with the intention of regularising estimation while allowing the data to dominate posterior inference. Prior perturbation techniques are assumed to be proportional to the magnitude of the perturbation.

Limitations Limitations include the heavy computational challenge in MCMC techniques limiting the full range of values that can be tested. Moreover, findings are limited to the current set of evaluation techniques being used and may miss characteristics exhibited by resulting posterior distributions. Furthermore, the robustness conclusions drawn are conditional on the assumed Autoregressive structure, including Student-t distribution innovations and first-order Autoregressive dynamics for latent volatility and regime processes; misspecification of these structural assumptions may induce posterior stability that reflects model rigidity rather than genuine market behaviour.

1.4 Relevance to Quantitative Finance

Determining the robustness of posterior distributions obtained from stochastic processes is an increasingly important issue in modelling financial data. The issue is a long standing one; Broto, Ruiz (2004) asserts that the empirical application of stochastic volatility models may be limited as estimation of parameters can prove to be difficult or unreliable in their literature review. However, the importance of including stochastic processes in modelling financial

data is established by Cunchala (2024), exploring a variety of different stochastic approaches in financial market modelling and establishes its effectiveness in capturing market dynamics.

Early Bayesian stochastic volatility models, such as Kim et al. (1998), demonstrate that simulation-based Bayesian inference can effectively correct for approximation errors arising from linearised likelihood representations. Although this work represents a significant methodological advance, posterior distributions are currently largely treated as reliable once convergence is achieved, with limited examination of their robustness to prior specification or structural modeling choices. As a result, the extent to which inferred volatility dynamics and parameter estimates are driven by data versus modelling assumptions remains unclear, particularly when comparing alternative specifications.

This ambiguity is consequential in the context of hierarchical latent-variable models. Roos et al. (2015) highlight that Bayesian hierarchical models are especially prone to weak identifiability and overparameterisation, such that posterior concentration may obscure underlying structural uncertainty. In such settings, it may be difficult to determine which parameters are well identified by the data and which are primarily influenced by prior assumptions. Together, these observations suggest that while early Bayesian stochastic volatility models establish feasibility and computational effectiveness, they provide limited guidance on assessing the reliability of posterior inferences for economic interpretation or decision-making.

While the importance of posterior robustness is increasingly acknowledged in the Bayesian literature, systematic sensitivity analysis remains difficult to implement in practice, particularly for hierarchical models with latent stochastic processes. Müller (2013) demonstrates that posterior quantities of interest can exhibit pronounced sensitivity to local perturbations in prior specification, with such effects propagating non-linearly through high-dimensional parameter spaces. These challenges are amplified in models with latent states, where identifiability is often weak and posterior dependence structures are complex.

In applied settings, this complexity has limited the routine adoption of sensitivity analysis. Roos et al. (2015) additionally shows that hierarchical Bayesian models may exhibit overparameterisation and partial identifiability, such that posterior concentration alone is insufficient to establish inferential reliability. From a computational perspective, practical sensitivity analysis is further constrained by the cost of repeated posterior evaluation under alternative priors or structural assumptions. Modelling with Autoregressive Conditional Heteroskedasticity techniques and changes in regimes by Hamilton and Susmel (1994) emphasizes that these limitations are particularly relevant in time-series models with regime-switching and stochastic volatility components, where inference about persistence and regime dynamics may be driven as much by modelling assumptions as by the observed data. Consequently, despite its recognised importance, Jacobi, Joshi and Zhu (2018) notes that posterior sensitivity analysis remains non-standard in empirical volatility modelling, creating a gap between methodological capability and applied practice.

This study seeks to contribute to Bayesian posterior distribution analysis in volatility modelling to not only contribute to understanding of which posterior distributions are reliable or not within a AR(1) context, but also in which periods and contexts. Understanding the reasoning and appropriate usage of obtained posterior distributions will allow more accurate applications of such models across methods within volatility modelling.

2 Computational and Numerical Framework

The objective of this study is to analyse the posterior distributions obtained by NUTS from an AR(1) model and the proceeding inferred regimes by evaluating its the robustness and reliability. Due to the difficulty in dealing with intractable stochastic likelihoods, prior perturbations formalised by Weiss (1996) are applied to test this. Perturbed posterior distributions are analysed via KL Divergence and ELPD to investigate and draw conclusions regarding the robustness of the posterior distributions.

A first order autorgressive model that features additive fast–slow latent volatility components is jointly analysed with inferred regimes based on the inferred NUTS volatility via a hidden markov model. This modular approach

provides a controlled setting in which posterior robustness can be examined under increasing model complexity rather than arising from a single model-specific artefact. Observed returns are conditionally t-distributed with volatility driven by latent autoregressive processes, where Bayesian inference via NUTS targets the joint posterior distribution over model parameters and latent volatility states. Given the estimation difficulties of stochastic likelihoods, as established by Broto and Ruiz (2004), we evaluate the sensitivity of the posterior distribution by varying prior's to observe any changes in the resulting posterior (Gelman et al., 2013, p.436). Techniques formalised by Weiss (1996) are implemented to assess how the established baseline joint posterior distribution induced by the AR(1) model as outlined in Equation 1 performs under varied prior distributions, where $r_{1:T} = (r_1, \ldots, r_T)$ denote observed returns, θ denotes model parameters, $h_{1:T}$ latent log-volatility states, and $z_{1:T}$ latent regime indicators.

$$p(\theta, h_{1:T}, z_{1:T} \mid r_{1:T}) \;\propto\; p(r_{1:T} \mid h_{1:T}, z_{1:T}, \theta)\, p(h_{1:T} \mid z_{1:T}, \theta)\, p(z_{1:T} \mid \theta)\, p(\theta), \tag{1}$$

Assessing perturbed posterior distributions using KL Divergence, which quantifies distributional robustness, together with ELPD, which evaluates predictive adequacy, allows posterior reliability to be assessed both in terms of inferential stability and economic usefulness. Examining these measures across differing market regimes and prior perturbations distinguishes between posterior behaviour driven by volatility dynamics or from modelling artefacts.

2.1 Perturbation Framework

Given the baseline joint posterior distribution is analytically intractable and approximated via NUTS sampling, following Weiss (1996), posterior perturbations are introduced through a nonnegative multiplicative perturbation function $h^*(\theta, h_{1:T}, z_{1:T}; \omega)$, where ω indexes the perturbation type and magnitude. The perturbed posterior following Weiss (1996) is defined as

$$p_\omega(\theta, h_{1:T}, z_{1:T} \mid r_{1:T}) = \frac{h^*(\theta, h_{1:T}, z_{1:T}; \omega)\, p_0(\theta, h_{1:T}, z_{1:T} \mid r_{1:T})}{\int h^*(\theta, h_{1:T}, z_{1:T}; \omega)\, p_0(\theta, h_{1:T}, z_{1:T} \mid r_{1:T})\, d\theta\, dh_{1:T}\, dz_{1:T}}. \tag{2}$$

By reweighting the baseline posterior through the perturbation function h^* the resulting posterior reflects the impact of modifying either the data contribution, the prior specification, or the observed return values, depending on the chosen perturbation. We adopt a combined global and local perturbation framework—where prior perturbations and case deletion act globally with observation-value shifts act locally.

Case Deletion Perturbation A case deletion perturbation assesses model sensitivity by systematically removing individual observations and examining the resulting change in posterior inference as it removes its likelihood contribution, allowing the influence of that observation on the posterior to be isolated. Consequently, this perturbation yields a posterior distribution conditional on $r_{1:T\setminus t}$, corresponding to inference under the removal of observation r_t. In this context, case deletion is particularly informative for assessing posterior reliability in modelling volatility dynamics, shown by Galea et al. (2025) in their diagnostic analysis for the stochastic Vasicek Model, as influential observations often coincide with volatility events. Extreme returns occurring during volatility clustering or near regime boundaries may disproportionately affect inferred persistence parameters, latent volatility paths $h_t^{(f)}$ and $h_t^{(s)}$, or regime allocations z_t.

Prior Perturbation Another global perturbation method, prior perturbations is implemented which is able to assess the sensitivity of the posterior distribution to varying priors & the sampling model, as established by Lavine (1991). The resulting in the perturbed posterior after applying a prior perturbation is:

$$p_q(\theta, h_{1:T}, z_{1:T} \mid r_{1:T}) \;\propto\; p(r_{1:T} \mid h_{1:T}, z_{1:T}, \theta)\, p(h_{1:T} \mid z_{1:T}, \theta)\, p(z_{1:T} \mid \theta)\, q(\theta). \tag{3}$$

This result is particularly relevant given the weak identification of persistence parameters, and latent volatility dynamics in finite samples. Stability of posterior inference under alternative priors provides evidence that key volatility indicators—such as persistence, clustering behaviour, and regime structure—are primarily data-driven, whereas substantial changes in posterior predictive performance indicate reliance on prior assumptions to regularise inference. As shown by Weiss (1996), perturbations defined through density-ratio reweighting are sufficient in that all inferential changes induced by modifying the model are transmitted through the resulting perturbed posterior, justifying posterior-based diagnostics such as KL Divergence & ELPD for robustness assessment.

Observation-Value Perturbation To assess sensitivity to local data perturbations, a single return r_t is shifted to $r_t + \delta$. This yields a posterior corresponding to the model in which only observation t is locally perturbed. Local observation perturbations, documented by Millar & Stewart (2007), provides an assessment of the sensitivity of the posterior distributions to weighted changes to an observation. Following this, in line with Wang (2012) who documents varying posterior distributions effects on model parameters and latent volatility, the stability of the latent volatility parameters and inferred regime in the implemented AR(1) model can be assessed if observation-value perturbations lead to local changes around t and minor adjustments in nearby latent volatility paths, or if there are large changes in persistence, clustering and mean reversion, heavy-tailed return behaviour.

2.2 Measures — KL Divergence & ELPD

KL Divergence quantifies how a probability distribution Q diverges from an expected probability distribution P, measuring the expected log-density loss incurred when Q is used in place of P. The divergence is measured in *nats* — units of information corresponding to natural logarithms — where one nat is equal to one unit of information under the natural logarithm. Intuitively, a KL value of x nats implies an average multiplicative density distortion of approximately e^x between the two distributions. As such, KL serves as a natural Bayesian loss function, capturing the expected additional log-loss under distributional misspecification. Applying this framework allows us to quantify how much information is lost when comparing our Bayesian posterior predictive return distribution to empirical market return distributions, as demonstrated by Zhu et al. (2012). As we are working in discrete time, the relevant KLD between two Gaussian approximations is defined as:

$$\mathrm{KL}\big(\mathcal{N}(\mu_p, \sigma_p^2) \,\big\|\, \mathcal{N}(\mu_q, \sigma_q^2)\big) = \log \frac{\sigma_q}{\sigma_p} + \frac{\sigma_p^2 + (\mu_p - \mu_q)^2}{2\sigma_q^2} - \frac{1}{2}. \tag{4}$$

We can empirically determine the sensitivity of our model's distribution to alternative priors to test for model robustness, thereby quantifying the extent to which perturbations alter the information content and density structure of inferred volatility dynamics. At its foundations however, Pettit (1986), Weiss and Cook (1992) show how typical divergence measures, such as assessing the influence of y_i by comparing $p(\tau \mid Y)$ and $p(\tau \mid Y_{(i)})$, are limited due to its scalar nature, showing a difference if present but withholds any additional information, motivating the use of ELPD.

ELPD is a measure of prediction accuracy of a model, a key metric in Bayesian statistics. By applying this measure to unperturbed samples of the model posterior distribution compared to perturbed samples, we can assess specific cases the AR(1) model begins to vary in its volatility modelling and evaluate robustness. For a given observation y_t,

$$\mathrm{ELPD}_t = \log \int p(y_t \mid \theta)\, p(\theta \mid y_{-t})\, d\theta, \tag{5}$$

where $p(\theta \mid y_{-t})$ denotes the posterior distribution obtained under the removal of observation y_t (approximated via Pareto-smoothed importance sampling. To assess robustness, we compare this quantity to the corresponding predictive density under the unperturbed posterior, yielding the ELPD difference

$$\Delta\mathrm{ELPD}_t = \mathrm{ELPD}_t^{(\mathrm{pert})} - \mathrm{ELPD}_t^{(\mathrm{base})}. \tag{6}$$

Large negative values of ΔELPD_t indicate observations or regimes for which the volatility dynamics implied by the AR(1) model and inferred regimes volatility dynamics and regime inference are particularly sensitive, thereby signalling reduced posterior reliability.

2.3 Methodology and Numerical Approach

An Autoregressive model featuring an additive fast-slow stochastic volatility component is implemented, a considerable pivot from the progress check report as the posterior distribution geometry created by marginalised regimes combined with an autoregressive stochastic volatility component proved to be too complex for a standard implementation of NUTS to explore. Posterior inference for such models is computationally challenging due to the presence of high-dimensional latent volatility states $h_{1:T}$ and discrete regime indicators $z_{1:T}$, yielding an analytically intractable and non-differentiable likelihood if sampled directly over regimes.

First-order Autoregressive Model Log-volatility is decomposed into two additive autoregressive components representing long-run (slow) and short-run (fast) volatility dynamics. Returns evolve according to Student-t distribution

$$r_t \mid h_t \sim \text{Student-}t(\nu, \mu_r, \exp(h_t/2)), \tag{7}$$

where $\nu > 2$ denotes the degrees of freedom parameter governing tail thickness, and total log-volatility is defined as $h_t = h_t^{(s)} + h_t^{(f)}$ - as heavy tails being a stylised feature of financial returns, the Student-t distribution avoids misattributing extreme returns to volatility persistence. The slow component captures persistent, long-horizon volatility behaviour and follows a stationary AR(1) process,

$$h_t^{(s)} = \mu_s + \phi_s \left(h_{t-1}^{(s)} - \mu_s\right) + \kappa_s \varepsilon_t^{(s)}, \tag{8}$$

while the fast component captures short-lived volatility shocks through

$$h_t^{(f)} = \mu_f + \phi_f \left(h_{t-1}^{(f)} - \mu_f\right) + \kappa_f \varepsilon_t^{(f)}. \tag{9}$$

Here, $\varepsilon_t^{(s)}$ and $\varepsilon_t^{(f)}$ denote independent standardised innovations, while κ_s and κ_f scale the magnitude of volatility-of-volatility shocks in each component. Persistence parameters ϕ_s and ϕ_f are assumed constant over time, ensuring stationarity and interpretability of the volatility decomposition. The Student-t observation equation accommodates excess kurtosis in returns while preserving conditional tractability given the latent volatility process.

Hidden Markov Model Regime Inference. Let $z_t \in \{1, \ldots, K\}$ denote a latent volatility regime at time t, evolving according to a first-order Markov chain with initial probabilities $\pi_j = \mathbb{P}(z_1 = j)$ and transition probabilities $P_{ij} = \mathbb{P}(z_t = j \mid z_{t-1} = i)$. Regime inference is performed *post hoc* on the inferred latent log-volatility sequence $\{h_t\}_{t=1}^T$, rather than being embedded directly within the stochastic volatility model for returns.

Following Bayesian estimation of the stochastic volatility model via NUTS, posterior draws of $h_{1:T}$ are treated as observed inputs to a Gaussian hidden Markov model, with regime-dependent emission densities $p(h_t \mid z_t = j)$.

For fixed HMM parameters, the discrete regime sequence $z_{1:T}$ is marginalised using the forward–backward algorithm, yielding smoothed regime probabilities $\gamma_{t,j} = \mathbb{P}(z_t = j \mid h_{1:T})$. Averaging these quantities across posterior draws propagates volatility uncertainty into regime inference, enabling interpretation of persistent low- and high-volatility regimes without introducing discrete latent states into the primary Bayesian model.

This recursion integrates out all regime paths without enumerating them, yielding a likelihood that is smooth in the continuous parameters and latent states $(h_{0:T}, \vartheta)$. Posterior inference therefore proceeds via NUTS, while regime summaries (e.g., $\gamma_{t,j} = \mathbb{P}(z_t = j \mid r_{1:T})$) are recovered post hoc from the forward–backward quantities.

Table 1: Posterior targets under baseline and perturbed specifications.

Target	**Posterior (up to proportionality)**
Case deletion (drop r_t)	$\pi_{-t}(\theta, h_{1:T} \mid r_{1:T\setminus t}) \propto p(r_{1:T\setminus t} \mid \theta, h_{1:T})\, p(\theta)\, p(h_{1:T} \mid \theta)$
Prior perturbation ($p \to p_\eta$)	$\pi_\eta(\theta, h_{1:T} \mid r_{1:T}) \propto p(r_{1:T} \mid \theta, h_{1:T})\, p_\eta(\theta)\, p(h_{1:T} \mid \theta)$
Observation-value shift ($r_t \to r_t + \delta$)	Let $r_{1:T}^{(t,\delta)}$ denote the perturbed dataset with $r_t^{(t,\delta)} = r_t + \delta$ and $r_s^{(t,\delta)} = r_s$ for $s \neq t$. Then $\pi_{t,\delta}(\theta, h_{1:T} \mid r_{1:T}^{(t,\delta)}) \propto p(r_{1:T}^{(t,\delta)} \mid \theta, h_{1:T})\, p(\theta)\, p(h_{1:T} \mid \theta)$

Posterior Inference via NUTS. The implementation of NUTS augments the target posterior with auxiliary momentum variables to define a Hamiltonian system whose stationary distribution corresponds to the desired posterior. Specifically, sampling is performed on the augmented density

$$p(\theta^*, r) \;\propto\; \exp\!\left(\log \pi(\theta^*) - \frac{1}{2} r^\top r\right), \qquad r \sim \mathcal{N}(0, I), \tag{10}$$

where $\theta^* = (\theta, h_{1:T})$ denotes the collection of continuous model parameters and latent states.

By exploiting gradient information of the log-posterior, NUTS efficiently explores the high-dimensional and highly correlated posterior distribution characteristic of first-order Autoregressive models, while avoiding the random-walk behaviour typical of conventional Metropolis–Hastings (Metropolis (1953), Hastings (1970)) algorithms. NUTS augments the Hamiltonian system by introducing a slice variable $u \mid (\theta, r) \sim \text{Uniform}\big(0,\ \exp\{\mathcal{L}(\theta) - \frac{1}{2} r^\top r\}\big)$, where $\mathcal{L}(\theta)$ denotes the log-posterior density and r is an auxiliary momentum variable. Conditional on u, the joint distribution of (θ, r) is uniform over the level set $\big\{(\theta, r) : \exp\big(\mathcal{L}(\theta) - \frac{1}{2} r^\top r\big) \geq u\big\}$.

Given the slice variable, NUTS simulates Hamiltonian dynamics both forward and backward in fictitious time using the leapfrog integrator. Trajectories are expanded recursively via a doubling procedure, implicitly constructing a balanced binary tree of candidate states until a *U-turn* condition is detected to avoid a retracement of steps. At termination, a proposal is sampled uniformly from the set of states generated during the trajectory, ensuring preservation of detailed balance and convergence to the target posterior distribution.

Prior Perturbations Implemented. Give the outlined NUTS framework, posterior sensitivity is assessed by applying controlled perturbations to the target posterior density while holding the numerical inference procedure fixed. Case deletion and observation-value perturbations modify the marginal likelihood computed via the forward algorithm, whereas prior perturbations modify the prior distribution. In all cases, the resulting perturbed posterior for each perturbation technique in Table 1 defines a new target density for NUTS in the following generic form:

$$\tilde{\pi}(\theta, h_{1:T} \mid \cdot) \;\propto\; \tilde{p}(r_{1:T} \mid \theta, h_{1:T})\, \tilde{p}(\theta)\, p(h_{1:T} \mid \theta), \tag{11}$$

where $(\tilde{p}(r_{1:T} \mid \theta, h_{1:T}), \tilde{p}(\theta))$ correspond respectively to the perturbed likelihood and/or perturbed prior under the chosen perturbation scheme. This allows differences in posterior behaviour to be attributed solely to the perturbation rather than changes in the inference algorithm.

2.4 Data Handling & Preprocessing

The required financial time series are Equity and Foreign Exchange data, sampled Daily, and will be sourced from data vendor Barchart.com, specifically high-volume equity assets to ensure sufficient market participation to analyse, entailing Lynas Rare Earths Ltd (LYC.AX), Commonwealth Bank of Australia (CBA.AX) and 4DMedical (4DX.AX) capturing varying market volatility characteristics unique to each respective sector. Contrastingly with the Foreign Exchange market, sourced currencies will be EURUSD, GBPJPY and USDJPY given their triangular-

denominated relationship with the USD and JPY common to all 3 pairs to assess if the model will show similar volatility characteristics.

Autoregressive and Hidden Markov Models assume stationarity; as raw financial time series data is often non-stationary, the logarithm of Box-Jenkins (1970) methodology of differencing is implemented in preprocessing the returns of the time series to obtain a stationary time series that still retains its characteristics.

2.5 Numerical Implementation

Volatility Modelling and Regime Inference Implementation Due to computational limitations, we implement posterior inference of only the latest $T = 500$ data points of each of our analysed *perturbed* time series where our baseline had $T = 2500$, with an AR(1) model using NUTS with 4 parallel chains, each comprising 1,500 warm-up iterations and 1,500 retained draws (3,000 warm-up iterations and retained draws for the baseline). The target acceptance rate is set to 0.97 and the maximum tree depth to 12 to mitigate divergent transitions in the high-dimensional latent-state posterior. Since Hamiltonian Monte Carlo requires differentiable targets, the stochastic volatility model is estimated independently of any discrete regime structure, yielding posterior draws of the latent log-volatility path $h_{1:T}$. Regime classification is then performed in a modular second stage. The posterior mean of $h_{1:T}$ is used to fit a Gaussian hidden Markov model with $K = 2$ states via expectation–maximisation, corresponding to low- and high-volatility regimes. The estimated transition matrix,

$$P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}, \qquad \sum_{j=1}^{2} P_{ij} = 1 \;\; \forall i, \tag{12}$$

is treated as fixed conditional on the fitted HMM. The forward–backward algorithm is applied to each posterior draw of $h_{1:T}$ under the fixed HMM parameters, producing smoothed probabilities $\gamma_{t,j} = \mathbb{P}(z_t = j \mid h_{1:T})$. Averaging these quantities across posterior draws yields regime probabilities that reflect both persistence in volatility dynamics and posterior uncertainty from the baseline AR(1) model, enabling robustness diagnostics on regime characteristics without introducing discrete latent states into the primary Bayesian sampler.

Perturbation Specifications Case deletion perturbations were implemented by taking the largest 3 absolute change cases in our data, and re-implementing a NUTS fit with for each case, where the observation shift was implemented by taking the standard deviation of the log diffed returns and multiplying it for each delta value in $[0.5, -0.5, 1.0, -1.0]$ and refitting NUTS accordingly. Prior Perturbations were implemented on the ϕ_s & ϕ_f prior's, and θ, where θ denotes $[\mu, \kappa, \phi]$ for all time series. Due to computation constraints, prior perturbations on tail parameter v values for both fast and slow components, was limited to 'CBA', '4DMedical', 'USDJPY' and 'GBPJPY'. Note that perturbation magnitudes were chosen to induce measurable but not destabilising posterior shifts.

Implementation of KL divergence and ELPD under perturbations For each perturbation setting $\omega \in \Omega$, we obtain posterior draws $\{(\theta^{(m)}, h_{1:T}^{(m)}, z_{1:T}^{(m)})\}_{m=1}^{M} \sim \tilde{\pi}_\omega$ using the same NUTS configuration to ensure comparability across runs. The posterior predictive return distribution is then approximated by $\tilde{p}_\omega(r) \approx \frac{1}{M}\sum_{m=1}^{M} p(r \mid \theta^{(m)}, h^{(m)}, z^{(m)})$. To compute a discrete KLD, we map both empirical returns and posterior predictive draws to a common finite support (e.g. histogram bins $\{b_k\}_{k=1}^{K}$), giving probability masses $\hat{P}(b_k)$ and $\hat{Q}_\omega(b_k)$, and evaluate $D_{\mathrm{KL}}(\hat{P} \,\|\, \hat{Q}_\omega)$.

Predictive performance is assessed via expected log pointwise predictive density (ELPD), computed using a leave-one-out approximation. For each observation r_t, we compare $\log p(r_t \mid r_{-t})$ under the baseline posterior and the perturbed posterior, summarising perturbation impact by $\Delta\mathrm{ELPD}(\omega)$. As case deletion changes the length of the dataset, normalisation is required in order to compare the perturbed $N-1$ length against the baseline N by implementing $\Delta\mathrm{ELPD} = \frac{elpd_{pert}}{N_{pert}} - \frac{elpd_{base}}{N_{base}}$ which is extended to the observation shift and prior pertubation calculations to keep ELPD calculations consistent.

3 Schematic / Algorithmic Representation

Figure 1 Enter Caption

Require: Observed returns $r_{1:T}$, stochastic volatility (SV) specification, baseline prior $p(\theta)$
Require: Prior perturbation set $\mathcal{W} = \{\text{scale_all},\ \text{persistence-scale},\ \text{tail-scale}\}$
Inputs:

- Observed return series $r_{1:T}$.
- Stochastic volatility (SV) specification: Additive fast–slow AR(1) latent log-volatility components, observation model $p(r_t \mid h_t, \theta)$ with Student-t innovations, Baseline prior distribution $p(\theta)$ over volatility parameters.
- NUTS configuration: $C = 4$, $N_{\text{warm}} = 1500$, $N_{\text{draws}} = 1500$, $\delta_{\text{acc}} = 0.97$, $D_{\max} = 12$.
- Number of HMM regimes $K = 2$.

Outputs:

- Baseline posterior draws $\{(\theta^{(m)}, h_{1:T}^{(m)})\}_{m=1}^{M}$ & Perturbed posterior draws $\{(\theta_{\omega}^{(m)}, h_{1:T,\omega}^{(m)})\}_{m=1}^{M}$, $\omega \in \mathcal{W}$.
- Robustness diagnostics: KL divergence and ΔELPD between baseline and perturbed posteriors.
- Posterior regime probability summaries $\bar{\gamma}_{t,k}$.

Preprocessing:
Difference log-price series to obtain stationary returns (Box–Jenkins methodology).

Construct the stochastic volatility (SV) model:
Define the observation model:

$$r_t \mid h_t \sim t_\nu(0,\ \exp(h_t/2)), \qquad \nu > 2.$$

Define additive fast–slow latent log-volatility: $h_t = h_t^{(s)} + h_t^{(f)}$.
Define AR(1) dynamics for each component $c \in \{s, f\}$:

$$h_t^{(c)} = \mu_c + \phi_c\Big(h_{t-1}^{(c)} - \mu_c\Big) + \sigma_c \varepsilon_t^{(c)}, \qquad \varepsilon_t^{(c)} \sim \mathcal{N}(0,1).$$

Assign weakly informative priors to $\theta = \{\mu_c, \phi_c, \sigma_c, \nu\}$.

Baseline inference:
Sample from the posterior $\pi_0(\theta, h_{1:T} \mid r_{1:T}) \propto p(r_{1:T} \mid h_{1:T}, \theta)\, p(h_{1:T} \mid \theta)\, p(\theta)$
using NUTS to obtain $\{(\theta^{(m)}, h_{1:T}^{(m)})\}_{m=1}^{M}$.

for each prior perturbation $\omega \in \mathcal{W}$ **do**
  Construct perturbed prior $\tilde{p}_\omega(\theta)$ via multiplicative scale perturbations.
  Sample from $\tilde{\pi}_\omega$ using the *same* NUTS configuration.
  Compute robustness diagnostics:
    – KL divergence between π_0 and $\tilde{\pi}_\omega$.
    – ΔELPD via leave-one-out predictive comparison.
end for

Post-hoc regime inference via Hidden Markov Model:
Extract S posterior draws $\{h_{1:T}^{(s)}\}_{s=1}^{S}$ and posterior mean path $\bar{h}_{1:T}$.
Fit a Gaussian HMM with K states to $\bar{h}_{1:T}$ using EM:

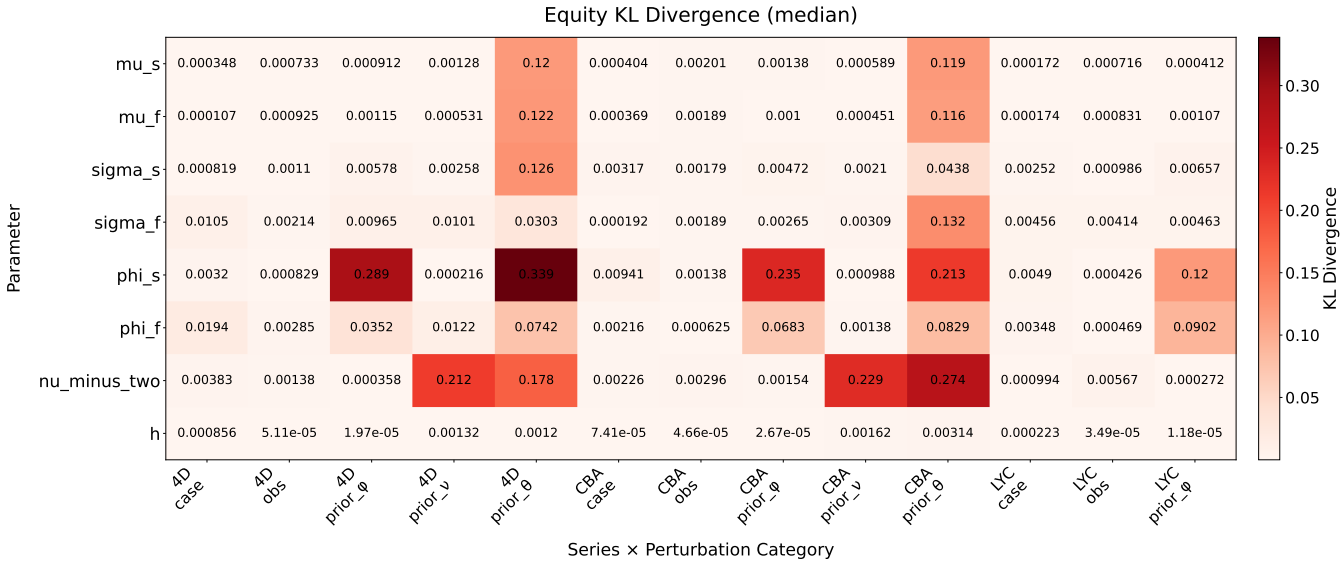
$$h_t \mid (z_t = k) \sim \mathcal{N}(m_k, v_k), \qquad \mathbb{P}(z_t = j \mid z_{t-1} = i) = A_{ij}.$$

Reorder regimes by increasing emission mean m_k.
for each draw $s = 1, \ldots, S$ **do**
  Apply forward–backward algorithm with fixed HMM parameters to obtain $\gamma_{t,k}^{(s)} = \mathbb{P}(z_t = k \mid h_{1:T}^{(s)})$.
end for
Aggregate regime probabilities: $\bar{\gamma}_{t,k} = \frac{1}{S}\sum_{s=1}^{S} \gamma_{t,k}^{(s)}$.

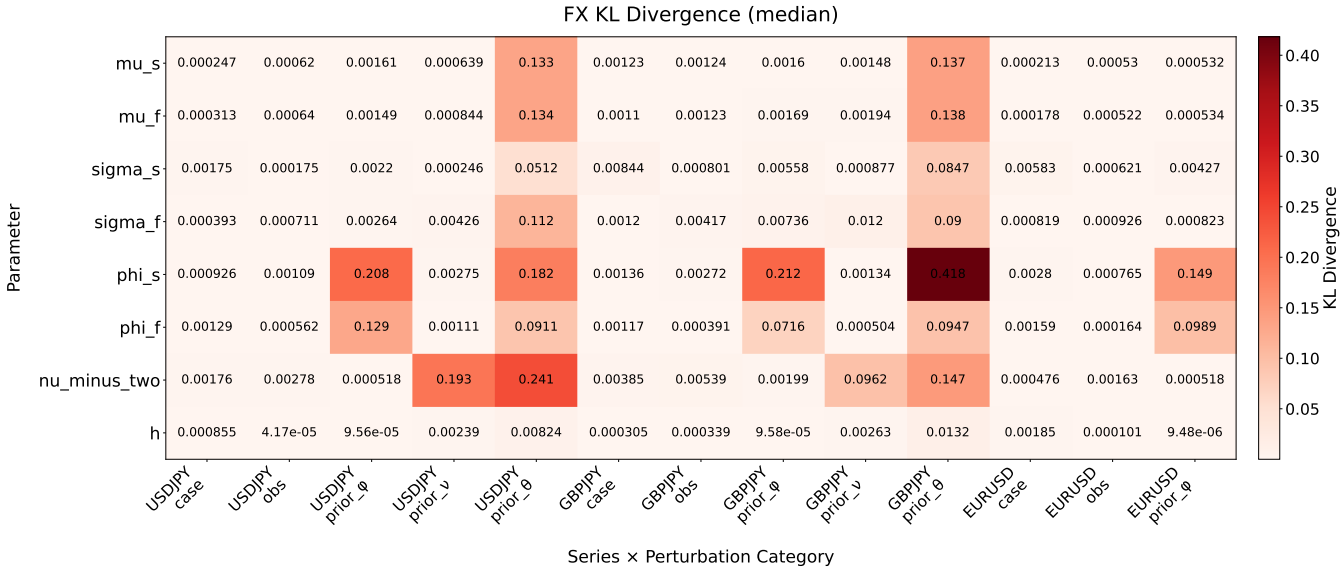
4 Results and Critical Analysis

4.1 Kullback-Leiber Divergence Results

The perturbations analysed via KL Divergence, as displayed in Figure 2's heatmap, with each value measured in nats as defined in Section 2.2, overall display magnitudes largely consistent with a robust model that captures mean reversion and volatility clustering, with some reservations regarding persistence sensitivity. As KL values closer to 0 imply closer alignment with the baseline posterior distribution P, the majority of perturbed posterior distributions Q remain close to P, suggesting that the inferred volatility dynamics are not artefacts of minor prior or data perturbations. A structural pattern emerges across both Equity and FX series: global prior perturbations applied to the full parameter vector $\theta = \{\mu, \kappa, \phi\}$ produce the largest KL shifts, with the strongest sensitivity observed in the slow-persistence parameter ϕ_s. The second largest shifts arise from perturbations applied to the tail parameter v, indicating non-trivial sensitivity of the posterior distribution to assumptions regarding tail thickness.



(a) Equities



(b) FX

Figure 2: KL divergence heatmaps: aggregated posterior sensitivity under perturbations.

Tail Prior Sensitivity. Perturbations applied to the Student-t tail parameter v reveal that posterior distributions are moderately sensitive to assumptions regarding heavy-tailed return behaviour. This is economically intuitive: in stochastic volatility models, tail thickness directly governs the attribution of extreme returns to either latent volatility spikes or heavy-tailed innovation structure. Altering v therefore shifts the balance between structural volatility dynamics and innovation-driven outliers. The non-negligible KL shifts observed under tail prior perturbations suggest that extreme observations contribute meaningfully to posterior geometry, though not in a manner that destabilises the model.

Importantly, this sensitivity remains controlled rather than explosive. While tail assumptions influence inference, the resulting posterior distributions remain within a bounded divergence range, implying that volatility clustering and persistence structure are not solely artefacts of distributional tail specification.

θ-Scale Perturbations and Persistence Sensitivity. The largest KL divergences are observed under global θ-scale perturbations, particularly affecting ϕ_s. This behaviour is consistent with known identification challenges in persistence parameters within AR(1) stochastic volatility settings. When ϕ_s approaches unity, small prior adjustments can meaningfully alter the implied half-life of volatility shocks, thereby producing amplified changes in posterior density structure.

This pattern suggests that the slow-moving volatility component is structurally central to the model's representation of clustering behaviour. Since ϕ_s governs long-run mean reversion, perturbations to the broader parameter vector θ disproportionately propagate through this channel. The resulting KL magnitudes therefore reflect structural sensitivity rather than numerical instability.

The most pronounced example occurs in GBPJPY under prior θ perturbation, where the KL value reaches 0.418 nats. Interpreting this magnitude, a KL divergence of 0.418 implies an average multiplicative density distortion of approximately $e^{0.418} \approx 1.52$, indicating that the perturbed posterior assigns densities roughly 52% different from the baseline distribution. Given GBPJPY's comparatively volatile nature, this sensitivity likely reflects genuine persistence instability rather than algorithmic weakness. In practical terms, this suggests that a ϕ_s estimate calibrated in one volatility regime (e.g., 2021–2022) may not generalise seamlessly across structurally distinct market conditions.

Case Deletion and Observation Shifts. Across both case deletion and observation-value perturbations, KL divergences remain uniformly small across parameters and latent volatility states. This indicates that removing individual extreme observations or locally shifting returns does not materially alter posterior inference. In particular, latent volatility $h_{1:T}$ remains stable under these perturbations.

If the AR(1)–NUTS framework were unrobust, case deletion—especially of the most volatile returns—would be expected to meaningfully impact σ parameters and influence the latent volatility paths. However, this effect is not observed but rather, volatility dynamics appear structurally absorbed by the autoregressive components rather than driven by isolated observations.

Similarly, observation-value shifts of δ produce negligible KL deviations. This suggests that the posterior geometry explored by NUTS remains smooth and stable under local likelihood modifications. The Student-t specification appears to successfully accommodate moderate return distortions without destabilising persistence or volatility-of-volatility parameters.

Latent Volatility Robustness. A key diagnostic question is what KL divergence would reveal if the model were unrobust. In hierarchical stochastic volatility settings, instability typically manifests first in latent state trajectories. However, the latent volatility parameter h consistently displays the lowest KL magnitudes across all perturbation classes.

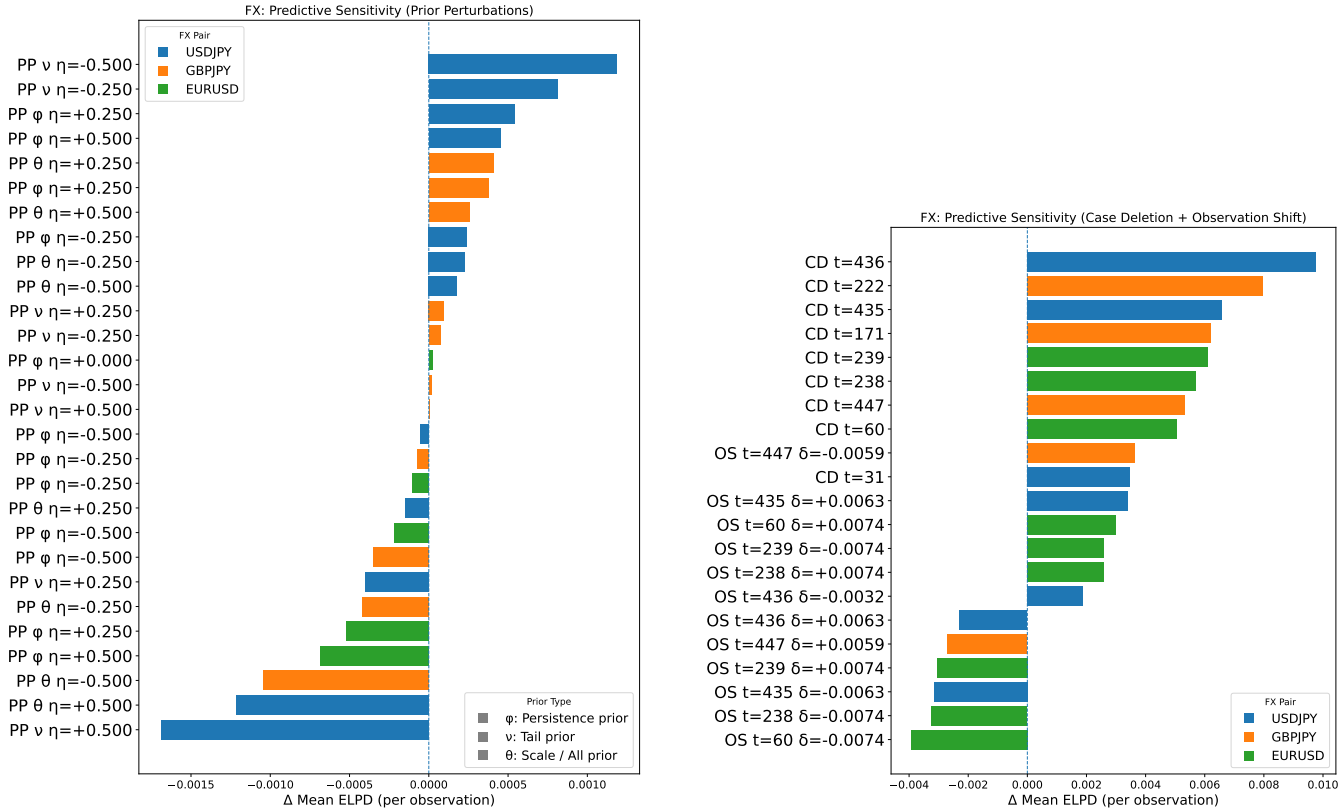
Even under the largest prior perturbations to θ, the distribution of $h_{1:T}$ remains tightly aligned with the baseline posterior. This suggests that while parameter-level posterior mass may shift moderately under altered priors, the

implied volatility path itself is structurally stable. Consequently, the AR(1)–NUTS framework appears capable of robustly recovering latent volatility dynamics even when persistence or tail parameters are perturbed.

Overall, the KL results indicate that posterior sensitivity is concentrated primarily in long-run persistence parameters rather than in latent volatility structure or regime dynamics. This supports the conclusion that the baseline posterior captures economically meaningful volatility features rather than reflecting sampling artefacts or weak prior regularisation. However, as KL divergence is inherently a scalar measure summarising distributional distance, it does not provide insight into predictive adequacy or observation-level sensitivity. To complement this distributional analysis, we therefore proceed to ELPD diagnostics to obtain a more granular assessment of predictive robustness under perturbation.

4.2 ELPD Results

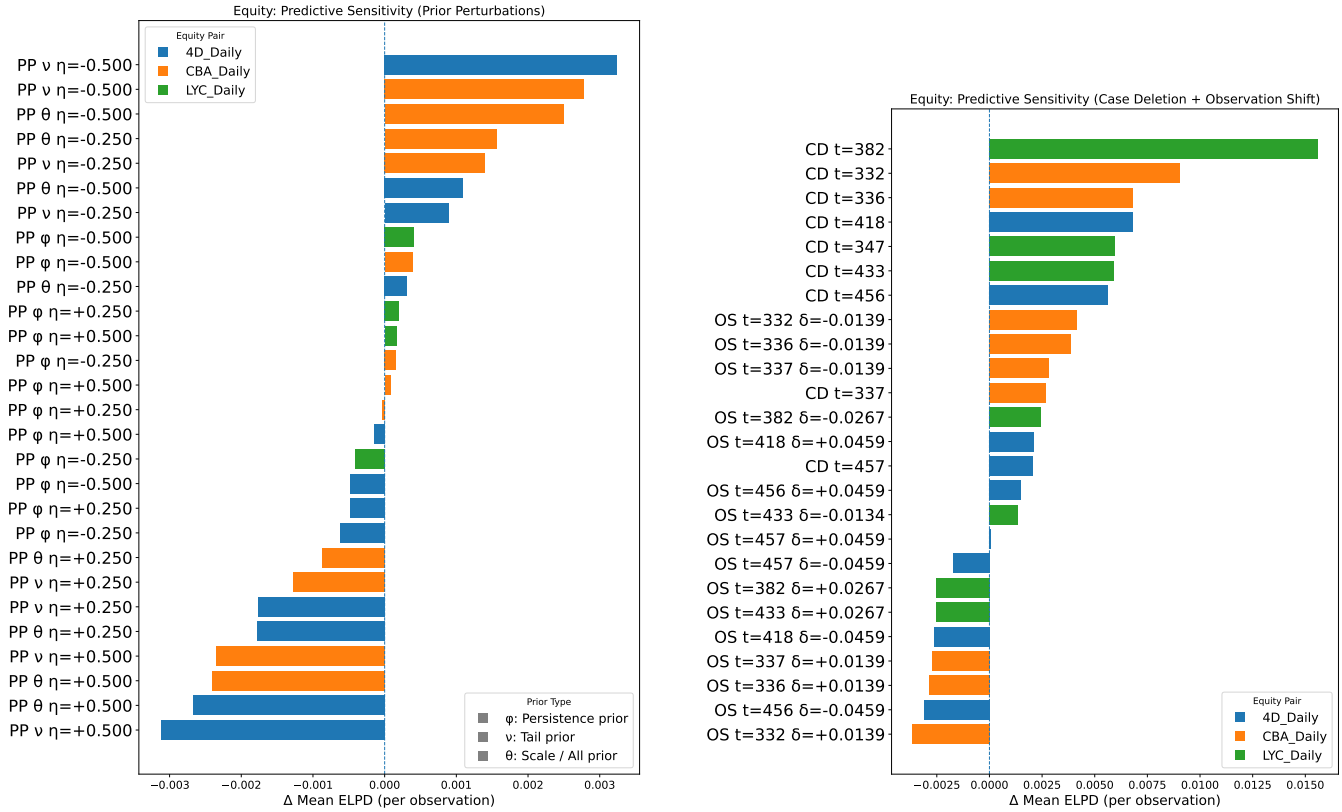
The ELPD analysis suggests strong predictive robustness of the baseline posterior distribution under the perturbations implemented. In several cases, perturbed specifications yield marginally positive ΔELPD values, indicating that the perturbed posterior assigns slightly higher expected log predictive density to held-out observations relative to the baseline- note that "predictive" refers to out-of-sample predictive density evaluation rather than forecasting in a time-series sense. This pattern is particularly evident across case deletion perturbations, where removal of influential observations occasionally improves predictive fit. Although the mean ΔELPD magnitudes are generally small which suggests robustness of the model —consistent with the KL divergence results and indicative of robustness—systematic trends remain economically informative.



(a) Prior Pertubations

(b) Case Deletion and Observation Shifts

Figure 3: Predictive sensitivity under perturbations (FX).



(a) Prior Perturbations

(b) Case Deletion and Observation Shifts

Figure 4: Predictive sensitivity under perturbations (Equity).

Case Deletions Improvements in mean ΔELPD under case deletion suggest that removing the most volatile timestamps marginally enhances predictive density fit. Since the deleted observations correspond to the largest absolute returns, this indicates that extreme volatility events exert local influence on predictive performance but do not fundamentally determine the structure of the latent volatility process.

This behaviour is economically intuitive. Although the Student-t specification is designed to accommodate heavy-tailed observations, extreme returns may still exert disproportionate leverage on short-run predictive density. The fact that predictive performance improves slightly upon their removal—particularly in more volatile assets such as LYC, GBPJPY and USDJPY—suggests that these observations introduce local predictive noise rather than structural regime shifts.

Crucially, the improvements remain small in magnitude. If posterior inference were primarily driven by a handful of extreme observations, case deletion would induce substantial predictive degradation. Instead, the limited ΔELPD shifts indicate that volatility clustering and persistence dynamics are embedded in the broader time series rather than being artefacts of isolated shocks. This supports the interpretation that the AR(1) volatility structure is capturing stylised behaviour robustly.

Tail Perturbation Implication Notably, perturbations applied to the tail parameter with $\delta = 0.500$ generate the largest negative shifts in ΔELPD, suggesting that predictive adequacy is most sensitive to assumptions governing heavy-tailed return behaviour. Since the tail parameter directly controls the allocation of probability mass to extreme returns, altering it meaningfully changes the model's predictive density in the tails.

This sensitivity highlights the economic importance of correctly specifying heavy-tailed innovation structure in stochastic volatility models. While KL divergence indicated bounded distributional shifts under tail perturbations,

ELPD reveals that predictive performance deteriorates more visibly when tail assumptions are altered. This suggests that tail behaviour contributes materially to predictive accuracy, even if the latent volatility path itself remains stable. In this sense, heavy-tailed modelling assumptions play a more critical role for predictive adequacy than for posterior geometry alone.

Observation Shifts and ϕ Prior Perturbations Across both Equity and FX series, observation-value shifts and ϕ prior perturbations generate ΔELPD values centred near zero, neither materially improving nor worsening predictive performance. Observation shifts modify individual timestamps locally, yet the negligible predictive impact indicates that the AR(1) latent volatility process absorbs such changes without materially altering out-of-sample density performance. Similarly, moderate perturbations to the persistence prior ϕ do not meaningfully affect predictive adequacy, despite producing observable KL divergence at the parameter level. This distinction is important: while persistence parameters may shift under prior scaling, the implied predictive distribution for returns remains stable.

If posterior distributions were artefacts of prior assumptions or regime instability, predictive performance would deteriorate under modest perturbations. Instead, predictive density remains largely invariant across local data modifications and persistence prior shifts. Sensitivity is concentrated primarily in tail specification rather than in volatility dynamics themselves, supporting the conclusion that stylised features—clustering and mean reversion—are captured structurally rather than being driven by fragile modelling assumptions.

4.3 Regime Analysis

The violin plots display the distribution of regime-level mean log-variance shifts between the perturbed posterior distributions and the baseline specification. Specifically, we compare the regime-specific average latent log variance under perturbation to the corresponding baseline value, yielding

$$\Delta\mu_k = \mathbb{E}[h_t \mid z_t = k]_{\text{pert}} - \mathbb{E}[h_t \mid z_t = k]_{\text{base}}.$$

Each violin therefore represents the distribution of these shifts across perturbation classes (Case Deletion, Observation Shift, and Prior), aggregated across posterior draws, where Regime 1 represents lower mean log-variance relative to Regime 2, where ordering is imposed post-estimation to ensure consistent economic interpretation across perturbation classes.

While this aggregation sacrifices granularity regarding individual prior perturbation types, it provides a clearer structural view of how perturbations affect inferred regime-level volatility magnitude. Table 2 complements this analysis by reporting median regime occupancy probabilities (π_1, π_2) and their shifts $(\Delta\pi_1, \Delta\pi_2)$ relative to baseline. The light blue bar in the violins represents the median shift, allowing structural comparison between perturbation classes.

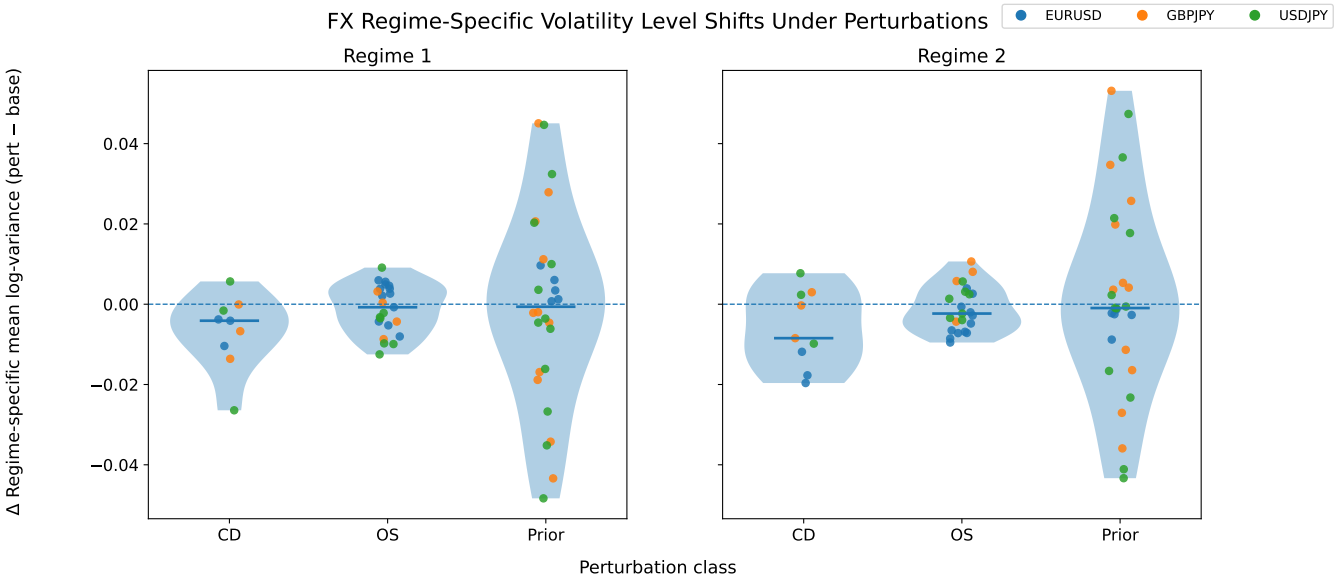
Observation Shifts. Observation-value perturbations produce regime-level shifts that are tightly centred around zero across both Equities and FX. Although minor shape differences are visible in the violin distributions, the regime-specific mean log variance remains nearly identical to the baseline specification. This indicates that local return distortions do not materially alter the latent volatility path nor the Gaussian HMM emission structure fitted to it. Given that the HMM assumes Gaussian emissions conditional on latent log variance, large regime instability would manifest as systematic shifts in regime means or crossing behaviour between regimes. The absence of such effects suggests that the regime interpretation remains stable under local data modifications. In this sense, the inferred regimes remain interpretable and economically consistent under observation shifts.

Case Deletions. Case deletion perturbations display slightly greater dispersion than observation shifts, except for Regime 2 of Equities. For Regime 1, the lower-volatility state, Equities exhibit an upward bias in $\Delta\mu_1$, largely driven

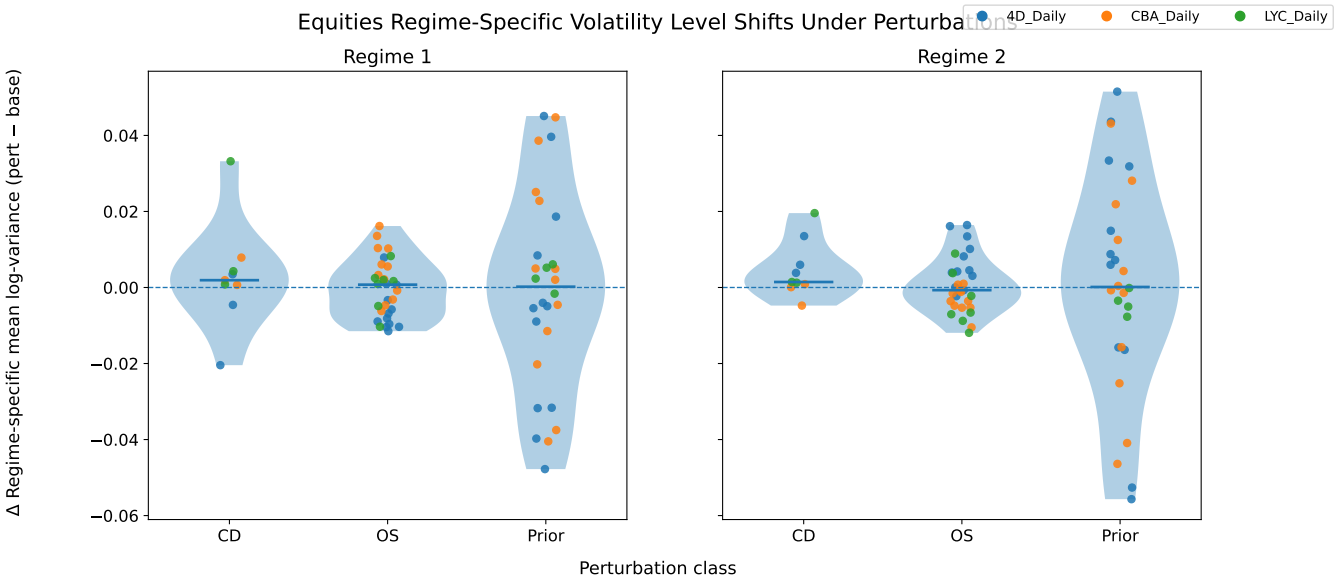
by LYC, whereas 4D appears slightly negative. However, these variations remain confined within a narrow range of approximately $[-0.025, 0.04]$ in log variance units. Considering Regime 1 and 2 for FX, we additionally see a narrow range of $[-0.025, 0.01]$, further establishing the marginal impact.

As h_t represents log variance, we can see that, albeit marginally, there is a tendency for FX to attribute a lower h to each of the regimes, whereas it's mixed for Equities. However, due to the limited T and assets of the study, this may be attributable to noise. Most importantly, these shifts do not induce structural reclassification of regimes. Table 2 further supports this conclusion, as regime occupancy probabilities remain stable across case deletions.

If the model were unrobust, removal of the most volatile timestamps would induce substantial reallocation of regime probabilities or collapse the separation between regime means. Instead, we observe only marginal adjustments, suggesting that volatility clustering is not driven by a handful of extreme observations but is embedded structurally within the time series.



(a) FX



(b) Equities

Figure 5: Regime-specific mean log-variance shifts under perturbations.

Prior Perturbations. The largest dispersion in regime-specific mean log variance is observed under prior perturbations, particularly for Regime 2, the high-volatility state. Although individual prior perturbation types are not disaggregated in the violin plot, the broader dispersion appears concentrated in assets such as GBPJPY, USDJPY, 4D, and CBA, whereas EURUSD and LYC remain more tightly centred around the median.

This pattern is consistent with earlier KL and ELPD findings indicating that θ-scale and tail (v) prior perturbations induce the most significant posterior shifts. Since ϕ_s governs long-run persistence and v governs heavy-tailed innovation structure, perturbing these parameters directly affects the amplitude and clustering of volatility spikes. The resulting dispersion in Regime 2 reflects sensitivity in the magnitude of high-volatility states rather than instability in regime allocation itself.

Dispersion in regime-specific volatility magnitude does not correspond to substantial shifts in regime probabilities. For example, USDJPY exhibits asymmetric regime occupancy probabilities ($\approx 0.33/0.67$), whereas GBPJPY is closer to balanced ($\approx 0.46/0.54$), yet both display comparable variance dispersion under prior perturbations. This suggests that regime magnitude sensitivity is driven primarily by structural parameter assumptions rather than regime frequency.

Structural Robustness Interpretation. Overall, regime-level shifts remain economically small and centred near zero across perturbation classes. Unrobustness would manifest as regime mean crossover, large asymmetric shifts, or substantial changes in occupancy probabilities. Instead, we observe stable regime separation and minimal probability drift (Table 2). Across all perturbation classes and series, no regime crossover is observed; the low-volatility state consistently maintains a lower mean log-variance than the high-volatility state. This preservation of regime ordering provides additional structural evidence that regime separation is not an artefact of prior specification or local data perturbation.

Therefore, while prior perturbations—particularly on θ and v—affect parameter-level posterior distributions, the structural interpretation of volatility regimes remains intact. This reinforces the central conclusion of the study: stylised features such as clustering and regime separation are captured robustly by the AR(1)–NUTS framework rather than being artefacts of fragile modelling assumptions. Although no out of sample regime forecasting is conducted, regime stability under perturbations validates the main question of the study which is the robustness of the model.

Table 2: Median regime occupancy probabilities and median shifts relative to baseline (Equities and FX).

Equities

Series	Pert.	π_1	π_2	$\Delta\pi_1$	$\Delta\pi_2$
4D	CD	0.6283	0.3717	0.0057	-0.0057
4D	OS	0.6180	0.3820	-0.0046	0.0046
4D	Prior	0.6186	0.3814	-0.0041	0.0041
CBA	CD	0.5402	0.4598	0.0000	0.0000
CBA	OS	0.5382	0.4618	-0.0020	0.0020
CBA	Prior	0.5385	0.4615	-0.0017	0.0017
LYC	CD	0.4677	0.5323	0.0014	-0.0014
LYC	OS	0.4656	0.5344	-0.0007	0.0007
LYC	Prior	0.4662	0.5338	-0.0001	0.0001

FX

Series	Pert.	π_1	π_2	$\Delta\pi_1$	$\Delta\pi_2$
EURUSD	CD	0.6127	0.3873	0.0057	-0.0057
EURUSD	OS	0.6072	0.3928	0.0002	-0.0002
EURUSD	Prior	0.6082	0.3918	0.0012	-0.0012
GBPJPY	CD	0.4775	0.5225	0.0140	-0.0140
GBPJPY	OS	0.4646	0.5354	0.0011	-0.0011
GBPJPY	Prior	0.4613	0.5387	-0.0022	0.0022
USDJPY	CD	0.3302	0.6698	0.0004	-0.0004
USDJPY	OS	0.3296	0.6704	-0.0002	0.0002
USDJPY	Prior	0.3302	0.6698	0.0004	-0.0004

4.4 Critical Analysis and Discussion

The results obtained across KL divergence, ELPD, and regime-level diagnostics collectively address the central research question: whether posterior distributions derived from the AR(1)–NUTS framework are artefacts of modelling assumptions or whether stylised volatility dynamics are captured robustly across perturbations.

From a distributional perspective, KL divergence indicates that posterior shifts remain bounded under all perturbation classes, with the largest deviations arising under global θ and tail (v) prior perturbations. This is

consistent with the theoretical framework of influence sufficiency (Weiss, 1996), wherein perturbations act through a sufficient function that governs posterior updating. In this context, the concentration of KL sensitivity in persistence and tail parameters reflects economically meaningful channels of influence rather than numerical instability. Importantly, latent volatility trajectories remain stable despite moderate parameter-level shifts.

Predictively, ELPD results reinforce this interpretation. While KL divergence captures global distributional differences, ELPD isolates predictive density stability. The small magnitude of ΔELPD across case deletions and observation shifts suggests that predictive adequacy is not materially altered by local perturbations. Sensitivity is most pronounced under tail prior modifications, indicating that heavy-tailed modelling assumptions play a central role in predictive performance. This divergence between KL and ELPD responses highlights an important distinction: distributional shifts do not necessarily translate into predictive degradation, underscoring the complementary nature of these diagnostics.

Structurally, regime-specific mean log-variance comparisons reveal minimal shifts in inferred volatility regimes under perturbation. Regime separation, magnitude, and occupancy probabilities remain stable (Table 2), suggesting that volatility clustering and mean reversion are embedded in the data rather than induced by fragile prior specifications. Were the model unrobust, one would expect regime crossover, substantial dispersion in regime means, or material shifts in regime probabilities under case deletion or prior perturbation. Such behaviour is not observed.

These findings align with the graphical influence philosophy of Weiss and Cook (1992), which cautions that scalar divergence measures may conceal structural nuance. By supplementing KL divergence with predictive and regime-level diagnostics, the analysis avoids overreliance on a single summary statistic. The convergence of evidence across distributional, predictive, and structural dimensions strengthens the robustness claim.

Nevertheless, limitations must be acknowledged. First, the HMM classification is applied post-estimation to the mean latent volatility path, rather than jointly estimated within the AR-HMM framework. While this separation simplifies interpretation, it abstracts from joint posterior uncertainty between regimes and volatility states. Second, KL divergence provides a scalar measure of global distributional distance and does not localise where in parameter space shifts occur. Third, importance reweighting and finite MCMC sampling introduce approximation error, particularly under larger prior perturbations where effective sample sizes may decline.

An unexpected outcome is that case deletion occasionally improves predictive density. This suggests that extreme observations, while accommodated by the Student-t likelihood, still exert mild leverage on predictive evaluation. However, the magnitude of these improvements is small and does not undermine regime stability, reinforcing the conclusion that volatility dynamics are not driven by isolated shocks.

Overall, the evidence indicates that posterior inference is most sensitive to structural assumptions regarding persistence and tail thickness, yet remains stable in its recovery of latent volatility paths and regime separation. The results, accompanied by satisfactory NUTS diagnostics of the perturbations ($\hat{r}$ scores of 1 to 1.01 and effective sample sizes with no divergences), therefore support the interpretation that stylised volatility features—clustering, mean reversion, and regime differentiation—are captured structurally by the AR(1)–NUTS framework rather than being artefacts of modelling assumptions or sampling instability.

Difficulties in Implementation A primary limitation encountered during implementation was computational timing, therefore tail and θ prior perturbations were restricted to four configurations rather than the full six initially proposed. While sufficient to identify structural sensitivity patterns, this constraint may limit the exhaustiveness and symmetry of the robustness analysis, particularly in assessing non-linear responses under more extreme prior scaling.

A second and more substantive difficulty arose in the initial attempt to jointly infer stochastic volatility dynamics and regime allocation within a unified AR-HMM NUTS framework. This joint specification produced substantial divergences and highly curved posterior geometries, consistent with the well-known challenges of sampling discrete regime indicators alongside persistent latent states despite marginalising. Given time constraints and convergence concerns, a modular approach was adopted: first estimating latent volatility via AR(1)–NUTS, followed by post-

estimation HMM regime classification.

While a fully joint regime-volatility specification would provide a more internally consistent robustness assessment—particularly in evaluating how perturbations propagate simultaneously through persistence and regime probabilities—the modular framework remains methodologically defensible. The latent volatility process is inferred independently of regime labels, and regime classification operates conditionally on this estimated structure. However, this separation may abstract from second-order posterior dependencies between AR parameters and regime allocation. Consequently, subtle feedback effects between persistence shifts and regime boundary identification may not be fully captured.

5 Critical Reflection and Conclusions

5.1 Methodological Assessment

While the perturbation framework implemented provides meaningful insight into posterior robustness, several methodological refinements could strengthen the analysis.

First, the case deletion and observation-shift perturbations could have been designed in a more dynamic and economically targeted manner. The current implementation focuses primarily on the most volatile timestamps and fixed-magnitude observation adjustments. A more structured perturbation scheme—for example, targeting clustered volatility periods, macro-announcement windows, or regime transition boundaries—may have yielded deeper insight into whether the AR(1)–NUTS framework remains stable under economically meaningful stress scenarios. Similarly, prior perturbations could have been more deliberately calibrated to answer time-specific structural questions, such as whether volatility persistence estimates calibrated during the March 2020 crash would generalise to calmer market conditions.

Computationally, NUTS proved highly efficient in exploring the posterior distribution of the AR(1) stochastic volatility model. Its adaptive step-size and mass-matrix tuning provide clear advantages over manually tuned Hamiltonian Monte Carlo, particularly in moderately high-dimensional latent state models. Relative to Gibbs sampling, NUTS avoids conditional conjugacy constraints and is able to traverse correlated parameter spaces more efficiently through gradient-informed proposals. However, alternative inference methods could have been considered. A Particle Monte Carlo (PMC) approach, particularly Particle Marginal Metropolis-Hastings (PMMH), may offer advantages in latent state models by jointly propagating state uncertainty through sequential importance sampling. In contrast to the modular approach adopted here, a particle-based method could more naturally accommodate state-space structure and potentially handle regime-switching dynamics without post-estimation separation. Similarly, for linear Gaussian state-space specifications, a Kalman filter would provide closed-form filtering and smoothing. Although the Student-t specification and nonlinear volatility structure prevent direct Kalman implementation, extended or unscented variants may offer approximate alternatives. These approaches could provide a useful benchmark for assessing whether robustness conclusions are sensitive to the sampling algorithm itself.

To further validate model assumptions and regime stability, an out-of-sample forecasting comparison between baseline and perturbed specifications would offer an additional dimension of evaluation. While KL divergence and ELPD assess distributional and predictive density robustness, a direct volatility forecasting exercise could reveal whether perturbations materially alter realised variance prediction accuracy. Even a simple rolling-window comparison would complement the current diagnostics and strengthen practical interpretability.

Finally, the time horizon considered—approximately 500 trading days—was partly constrained by computational feasibility. Although sufficient to capture multiple volatility episodes, longer or segmented windows (e.g., pre- and post-crisis subsamples) would allow more explicit testing of regime persistence across structurally distinct market environments. As such, the current analysis establishes a methodological precedent but does not exhaust the full space of robustness testing across time.

5.2 Conclusion

The objective of this research was to evaluate the robustness and reliability of posterior distributions obtained from an AR(1) stochastic volatility model estimated via NUTS, augmented with a regime overlay to assess the structural stability of inferred latent volatility. Specifically, the study sought to determine whether baseline posterior distributions were artefacts of modelling assumptions or whether stylised volatility features—clustering, mean reversion, and persistence—were captured structurally and robustly.

The results collectively support the latter conclusion. By perturbing priors and data through case deletion, observation shifts, and structured prior modifications grounded in Weiss' (1996) influence framework, new posterior distributions were obtained and systematically compared to the baseline using KL divergence, ELPD, and regime-level diagnostics. KL divergence demonstrated that distributional shifts remained bounded, with sensitivity concentrated primarily in persistence (ϕ) and tail (v) parameters. ELPD further showed that predictive adequacy remained stable under most perturbations, with only moderate sensitivity to heavy-tailed assumptions. Importantly, regime-specific mean log-variance comparisons revealed minimal structural change in volatility regimes, and regime occupancy probabilities remained stable across perturbation classes.

Taken together, these findings indicate that latent volatility paths are not mechanically driven by prior specification nor by isolated extreme observations. Rather, persistence and clustering behaviour emerge consistently across perturbations. The stylised facts of volatility—mean reversion in log variance, volatility clustering, and persistence—remain intact even when modelling assumptions are moderately altered. In this sense, the posterior distributions appear to reflect structural information in the data rather than artefacts of sampling geometry or prior calibration. As such, methodologically, the AR(1)–NUTS framework proved appropriate for the problem at hand as a sample model. NUTS provided efficient exploration of the posterior without requiring conjugacy assumptions, and the perturbation-based validation framework allowed for systematic stress-testing of modelling assumptions. The modular regime overlay, while not a fully joint AR-HMM specification, provided an interpretable and computationally feasible mechanism to assess regime stability. However, the inability to jointly infer regimes and volatility states represents a structural limitation, as subtle feedback effects between persistence parameters and regime probabilities may not be fully captured in the modular design.

This study also demonstrates that robustness validation in Bayesian stochastic volatility models can be operationalised using perturbation theory beyond purely theoretical settings. In quantitative finance applications—such as volatility forecasting, Value-at-Risk estimation, and derivative pricing—the reliability of posterior distributions is critical. If volatility estimates are highly sensitive to modest prior changes, downstream pricing and risk decisions may inherit instability. The bounded divergence and predictive stability observed here provide evidence that, at least within the scope of this model and sample, posterior-based volatility inference can be considered structurally reliable.

While the findings support robustness within the scope considered, they remain conditional on model structure, sample length, and modular regime classification. Should volatility dynamics exhibit structural breaks beyond the examined window, or should joint regime-volatility dependence materially alter posterior geometry, robustness conclusions may differ. Therefore, the conclusions should be interpreted as conditional robustness within the tested perturbation space rather than universal invariance. Despite this however, other constraints such as computational time and extent of prior perturbations, the framework developed establishes a scalable precedent for robustness validation in state-space models. Future extensions involving longer samples, crisis-period segmentation, joint regime-volatility estimation, or alternative volatility dynamics (e.g., Bayesian GARCH) would further strengthen generalisability.

In conclusion, the perturbation analysis indicates that the AR(1)–NUTS stochastic volatility framework captures stylised volatility features robustly across Equity and FX markets. The posterior distributions examined do not appear to be primarily artefacts of modelling assumptions but reflect structurally persistent dynamics in the underlying data-generating process.

5.3 Future Directions

Although the present implementation relies on relatively simple AR(1) stochastic volatility dynamics and modular regime classification, it establishes a framework for broader application of perturbation-based robustness analysis in Bayesian state-space modelling. There remains limited applied literature integrating graphical influence concepts with modern Bayesian stochastic volatility estimation, and this study contributes toward bridging that gap.

A natural extension would involve fully joint estimation of regime allocation and stochastic volatility within a unified AR-HMM framework estimated via NUTS or particle-based methods. Such an approach would allow perturbations to propagate simultaneously through persistence parameters and regime probabilities, providing a more granular view of structural stability. While computationally demanding, this extension would deepen insight into how regime boundaries themselves respond to prior and data perturbations.

Beyond AR(1) volatility, the framework could be extended to ARCH and GARCH-type models. Whereas AR(1) log-volatility models latent variance dynamics directly, GARCH models capture volatility clustering through deterministic variance recursion. Applying perturbation diagnostics to Bayesian GARCH specifications would allow comparison of robustness between latent-state and observation-driven volatility frameworks. Such comparison may reveal whether posterior sensitivity arises from latent state estimation or from variance recursion assumptions.

More broadly, the perturbation methodology has practical implications for risk management and derivative pricing. If posterior distributions used to estimate volatility surfaces or Value-at-Risk measures are sensitive to modest prior assumptions, downstream pricing and capital allocation decisions may inherit instability. Conversely, demonstrated robustness strengthens confidence in model-based volatility forecasts used in option pricing, stress testing, and portfolio allocation. Extending this framework to multi-asset portfolios or high-frequency volatility estimation could therefore provide meaningful applied contributions.

Overall, while computational and structural simplifications were necessary in the present study, the perturbation-based robustness framework developed here provides a scalable foundation for more complex Bayesian stochastic modelling applications.

6 GenAI Disclosure

I acknowledge the use of ChatGPT to draft and prepare Latex mathematical and plotting figures in the following sections:

- Section 2 — 2.5
- Section 4 — 4.1, 4.2, 4.3

Further use of ChatGPT was used for general phrasing recommendation for topic sentences in Section 4 and Section 5 in additional to clarity and grammer checks throughout the paper in addition to brainstorming of initial ideas to research.

Python scripts and Jupyter Notebooks were also constructed with the assistance of ChatGPT by explaining documentation from PyMC and giving example code blocks to begin to work off. Scripts were refactored accordingly to ensure understanding of the code.

Discussion of the meaning and implications of the results were also highlighted and explained by ChatGPT, measured against other known results and examples to ensure accuracy. Discussion of the "why's" and "what if's" were had in order to grasp the implications of the results. Consultation on how to navigate the numerous data files to analyse was also had.

Formatting of the Reference section was also generated by ChatGPT; note however it was only formatting and all resources was found by me. No information, data or sources were generated.

References

[1] Engle, R. F., and A. J. Patton (2001), "What good is a volatility model?", *Quantitative Finance*, 1(2), pp. 237–245.

[2] Broto, C., and E. Ruiz (2004), "Estimation methods for stochastic volatility models: A survey", *Journal of Economic Surveys*, 18(5), pp. 613–649.

[3] Müller, U. K. (2013), "Risk of Bayesian inference in misspecified models, and the sandwich covariance matrix", *Econometrica*, 81(5), pp. 1805–1849.

[4] Zhu, H., Ibrahim, J. G., and Tang, N. (2012), "Bayesian influence analysis", *Journal of the American Statistical Association*, 107(497), pp. 256–269.

[5] Pettit, L. I. (1986), "Diagnostic measures in Bayesian analysis", *Journal of the Royal Statistical Society, Series B*, 48(2), pp. 213–221.

[6] Weiss, R., and Cook, R. D. (1992), "A graphical case-statistic for assessing influence in regression", *Journal of the American Statistical Association*, 87(419), pp. 813–824.

[7] Weiss, R. E., and J. De La Horra (1996), "Sufficiency and influence", in *Bayesian Robustness*, IMS Lecture Notes–Monograph Series, Vol. 29, Institute of Mathematical Statistics, pp. 211–229.

[8] Gao, W., B. Wu, and B. Zhang (2024), "High-frequency volatility estimation and forecasting with a novel Bayesian GARCH–Itô model", *Electronic Journal of Statistics*, 18(2), pp. 3497–3534.

[9] Zhao, P., H. Zhu, W. Hung, and D. Lee (2024), "Neural network extensions of GARCH models for daily volatility forecasting", *arXiv preprint*, arXiv:2402.06642.

[10] Lavine, M. (1991), "Sensitivity in Bayesian statistics: The prior and the likelihood", *Journal of the American Statistical Association*, 86(414), pp. 396–399.

[11] Millar, R. B., and Stewart, W. S. (2007), "Assessment of locally influential observations in Bayesian models", *Bayesian Analysis*, 2(2), pp. 365–384.

[12] Wang, J. (2012), *Bayesian Estimation of Stochastic Volatility Models*, PhD Thesis, University of Sydney.

[13] Box, G. E. P., and Jenkins, G. M. (1970), *Time Series Analysis: Forecasting and Control*, Holden-Day, San Francisco.

[14] Metropolis, N., Rosenbluth, A. W., Rosenbluth, M. N., Teller, A. H., and Teller, E. (1953), "Equation of state calculations by fast computing machines", *Journal of Chemical Physics*, 21(6), pp. 1087–1092.

[15] Hastings, W. K. (1970), "Monte Carlo sampling methods using Markov chains and their applications", *Biometrika*, 57(1), pp. 97–109.

[16] Ferguson, T. S. (1973), "A Bayesian analysis of some nonparametric problems", *Annals of Statistics*, 1(2), pp. 209–230.

[17] Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013), *Bayesian Data Analysis*, 3rd ed., Chapman & Hall/CRC, Boca Raton.

[18] Gallo, G. M., and E. Otranto (2015), "Forecasting realized volatility with changing average levels", *International Journal of Forecasting*, 31(3), pp. 620–634.

[19] Hamilton, J. D., and R. Susmel (1994), "Autoregressive conditional heteroskedasticity and changes in regime", *Journal of Econometrics*, 64(1–2), pp. 307–333.

[20] Ding, Z., G. Kambouroudis, and D. McMillan (2025), "Forecasting volatility using regime-switching and heterogeneous autoregressive models", *Journal of Forecasting*, forthcoming.

[21] Cunchala, A. (2024), "Comparative analysis of stochastic volatility models: GARCH versus Heston", *Applied Mathematics and Computation*, 465, Article 127453.

[22] Jacobi, L., M. Joshi, and D. Zhu (2018), "Automated sensitivity analysis for Bayesian inference via Markov Chain Monte Carlo: Applications to Gibbs sampling", Working Paper, Department of Economics, University of Melbourne and Monash University.

[23] Galea, M., Molina, A., and Beaudry, I. S. (2025), "Bayesian stochastic volatility models with regime-switching dynamics", *Journal of Risk and Financial Management*, 18(2), Article 63.